

Frozen Lake from First Principles using Q-Learning

Technical Report — DSCD 614 Reinforcement Learning, Assignment 1

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1. Introduction

This report describes a complete Reinforcement Learning (RL) solution to the 8×8

2. Environment Design

The environment is implemented as the FrozenLakeEnv class with the required API

- State representation: single integer indices 0...63, where row = s // 8

The class maintains the current state, enforces boundaries, detects holes and

3. Q-Learning Algorithm

Q-Learning is a model-free, off-policy, value-based method. It maintains a

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$$

Here α is the learning rate, γ the discount factor, and

4. Training Methodology

The agent (QLearningAgent) is trained by train.py, which runs episodes,

Parameter	Value	Parameter	Value
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Hyperparameters were explored via command-line flags; $\alpha = 0.1$ with $\gamma = 0.99$ and a

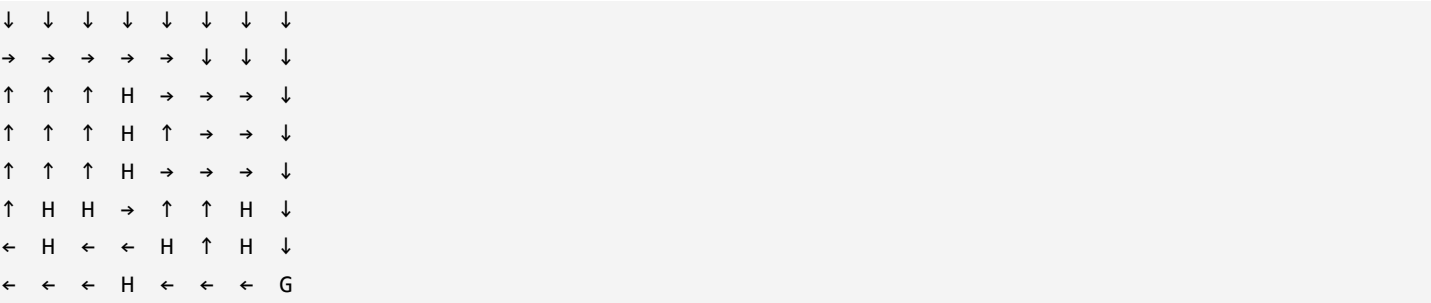
5. Experimental Results

Greedy evaluation over 100 episodes on the deterministic environment achieved

Bonus B (graphs): reward, success-rate and ϵ curves are generated

6. Learned Policy

The greedy policy extracted from the Q-table (Part D) avoids the central hole



Legend: ← Left, ↓ Down, → Right, ↑ Up, H Hole, G Goal. States that are off the

7. Challenges Encountered

- Reward sparsity: with reward only at the goal, early episodes rarely

8. Conclusion

A from-scratch tabular Q-Learning agent solves the deterministic 8×8 Frozen Lake

